

NY-ALESUND, Svalbard and Jan Mayen

WMO: 010070

Lat: 78.9228N

Lon: 11.9331E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-25.2	-22.7	-29.8	0.2	-24.2	-27.9	0.3	-21.9	18.1	-6.8	15.9	-6.4	2.3	130	0.937

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.0	9.9	6.8	8.7	6.1	7.9	5.6	7.3	9.3	6.5	8.4	5.8	7.6	4.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.8	5.7	7.9	4.9	5.4	7.2	4.1	5.1	6.6	23.3	9.4	21.6	8.5	20.3	7.7	12.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	12.4	10.6	-24.8	12.4	3.5	2.1	-27.2	14.0	-29.3	15.2	-31.2	16.4	-33.7	17.9
			-25.2	8.9	3.3	1.7	-27.5	10.2	-29.4	11.2	-31.3	12.1	-33.7	13.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.8	-9.2	-9.9	-11.7	-8.5	-2.0	2.9	5.9	4.6	1.3	-3.5	-6.9	-8.8
	DBStd	7.55	6.77	6.29	5.86	5.65	3.29	2.14	1.76	2.03	3.23	4.52	4.95	5.60
	HDD10.0	5028	594	556	671	556	373	212	127	168	262	420	507	583
	HDD18.3	8069	852	789	929	806	631	462	384	426	512	678	757	842
	CDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	3.9	4.8	4.8	4.5	3.8	3.5	3.1	2.9	2.9	3.5	4.0	4.6	5.0
(15) Precipitation	PrecAvg	473	60	41	46	29	18	16	29	38	58	41	53	45
	PrecMax	749	217	124	133	107	68	55	86	100	136	136	287	115
	PrecMin	269	6	3	10	11	0	3	2	4	8	1	6	5
	PrecStd	118	51	33	30	20	13	12	20	23	38	33	54	28
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	3.5	2.6	3.8	5.5	8.9	12.8	11.4	9.0	6.7	4.2	5.0
		MCWB	1.9	1.4	0.5	1.8	2.2	5.8	9.4	7.4	7.2	4.7	2.2	2.9
	2%	DB	2.0	2.0	0.7	2.2	3.7	7.7	10.6	9.3	7.6	5.2	2.7	1.9
		MCWB	0.4	0.4	-0.8	0.6	1.5	5.0	7.4	6.5	5.6	3.4	1.1	0.1
	5%	DB	0.8	0.3	-0.9	0.7	2.8	6.7	9.3	8.2	6.6	4.0	1.5	0.6
		MCWB	-0.6	-1.0	-2.1	-0.7	0.8	4.3	6.4	5.8	4.7	2.3	0.1	-0.9
	10%	DB	-0.5	-1.2	-3.0	-0.6	2.0	5.9	8.3	7.4	5.7	2.8	0.1	-0.8
		MCWB	-1.9	-2.5	-4.1	-1.9	0.1	3.6	5.9	5.2	4.0	1.4	-1.2	-2.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	1.7	0.9	2.1	2.7	6.4	9.8	7.9	7.4	5.1	2.7	3.0
		MCDB	3.5	3.2	2.3	3.6	4.3	8.3	12.2	10.1	8.6	6.5	4.0	5.1
	2%	WB	0.6	0.5	-0.7	0.7	1.8	5.3	7.8	6.9	5.9	3.7	1.3	0.3
		MCDB	1.9	1.8	0.7	2.0	3.5	7.2	9.8	9.0	7.3	5.0	2.6	1.6
	5%	WB	-0.6	-0.9	-2.2	-0.5	1.0	4.6	6.8	6.1	4.9	2.5	0.2	-0.7
		MCDB	0.7	0.2	-1.0	0.7	2.6	6.4	8.7	8.0	6.3	3.8	1.4	0.5
	10%	WB	-1.8	-2.4	-4.0	-1.8	0.2	3.9	6.1	5.4	4.1	1.5	-1.1	-2.0
		MCDB	-0.6	-1.2	-3.1	-0.6	1.8	5.5	8.0	7.1	5.6	2.8	0.0	-0.7
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.3	5.3	5.3	3.8	2.8	3.0	2.9	3.3	4.0	4.6	5.0
		MCDBR	5.3	5.8	6.3	4.8	4.0	3.8	4.2	4.0	4.3	4.3	5.3	5.8
	5% WB	MCWBR	4.7	5.3	5.7	4.1	2.8	2.5	2.8	2.6	3.5	3.6	4.7	5.3
		MCDWR	5.2	6.2	6.3	5.0	3.8	3.5	4.0	3.8	4.2	4.4	5.7	6.0
(40) Clear-Sky Solar Irradiance	taub		N/A	N/A	0.250	0.297	0.315	0.306	0.308	0.294	0.243	N/A	N/A	N/A
	taud		N/A	N/A	2.038	2.132	2.174	2.293	2.363	2.367	2.272	N/A	N/A	N/A
	Ebn at Noon		0	0	523	729	795	833	813	751	585	0	0	0
	Edh at Noon		0	0	49	85	100	94	83	68	39	0	0	0
(44) All-Sky Solar Radiation	RadAvg		0.00	0.00	0.64	2.54	4.49	5.47	4.50	2.84	0.99	0.00	0.00	0.00
	RadStd		0.00	0.00	0.06	0.26	0.38	0.44	0.37	0.23	0.10	0.00	0.00	0.00

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+1.52	+4.63	+4.10	N/A	N/A	N/A	-552	-549	N/A